TeraMove: Context-Aware Reconfiguration of THz Access Points for Resilient Pervasive Connectivity

***Abstract*—The Terahertz (THz) band promises high-speed indoor connectivity for 6G, but its dependence on line-of-sight makes it highly vulnerable to dynamic blockages. Traditional indoor deployments rely on static Access Points (APs) and cannot adapt when physical obstructions arise or crowd patterns shift, leading to low coverage. In this paper, we present TeraMove, a blockage-aware framework that uses spatial human-density context to detect THz link degradation from dynamic blockages and adapt the AP configuration to restore pervasive connectivity. We show an end-to-end process where this contextual information is reliably obtained through 3D LiDAR depth sensing, while our evaluation uses an empirical pedestrian density model. TeraMove first establishes an optimal baseline layout for AP placement using a Mixed Integer Linear Programming (MILP) model over 3D coordinates and beam orientations, then employs a four-tier optimization that escalates from electronic beam steering to full 3D node repositioning, performing physical movement only when lighter adaptations prove insufficient to provide threshold coverage. Evaluations in a realistic indoor environment show that crowd-induced blockages can reduce coverage to 16.6%; our algorithm recovers it to over 95% while requiring 65% less physical displacement than the alternative approaches evaluated.**

***Index Terms*—THz Communication, Context-Aware Adaptation, Pervasive Connectivity, Spatio-Temporal Blockage Modeling, Adaptive Infrastructure, HPSO, Dynamic Repositioning**

I. Introduction

The emergence of 6G networks is driven by a new class of applications, holographic communication, immersive extended reality, and digital twins, that demand sustained multi-Terabit-per-second data rates with ultra-low latency, far exceeding what current 5G infrastructure can deliver. The Terahertz (THz) band (0.1–10 THz), with orders of magnitude more available spectrum than sub-6 GHz or millimeter-wave (mmWave) bands, is widely regarded as the enabling physical layer for this next generational leap. However, THz signals suffer extreme path loss and molecular absorption, forcing transmitters to use narrow, highly directional beams that require an unobstructed Line-of-Sight (LoS) to every user. In populated indoor spaces, this LoS dependence is inherently fragile: a single human body crossing the beam can cause attenuation exceeding 20 dB, enough to collapse the link entirely.

The straightforward solution is static placement optimization, fixing transmitting nodes at predetermined positions to maximize coverage of the people in the building. Existing methods typically treat this as a 2D problem at a uniform ceiling height. While adequate for an empty floor plan, such deployments assume the propagation environment never changes. However, dynamic blockages caused by human mobility continuously reshape the interference landscape, and a layout providing near-perfect user coverage at one moment can suffer severe degradation when a crowd clusters unexpectedly. Depth sensing offers a highly reliable means to obtain such changing environmental context. Recent advancements in solid-state 3D LiDAR have shown that accurate human tracking and counting in indoor environments [**?**], [**?**]. Such sensing can provide the spatial human-density context needed to reason about changing blockage conditions. Static over-provisioning cannot anticipate every future crowd pattern and can lead to very high costs.

We present *TeraMove*, a context-aware framework that treats transmitting node deployment as a continuous adaptation problem rather than a one-time placement. TeraMove uses a spatial human-density representation to characterize dynamic blockage and first applies a Mixed-Integer Linear Programming (MILP) formulation to minimize the number of deployed transmitting nodes while satisfying coverage and SINR constraints. When crowd shifts degrade user coverage below a threshold, it triggers an Hierarchical Particle Swarm Optimization (HPSO) engine with a 4-tier cascading architecture: Tier 1 checks whether action is needed; Tier 2 electronically adjusts beam azimuth and beamwidth at zero physical cost; Tier 3 permits minor height changes; Tier 4, as a last resort, enables full 3D repositioning. Tier 2 alone recovers the majority of lost user coverage in most scenarios. A Hungarian algorithm-based assignment further minimizes aggregate physical displacement when movement is unavoidable.

Although focused on THz, the framework is transmitter-agnostic and applies equally to mmWave small cells, Wi-Fi access points, or 5G base stations, anywhere LoS-dependent links face dynamic blockages. We use the term "Access Points" (APs) throughout, noting the methodology extends naturally across deployment paradigms.

The main **contributions** of this paper are:

- A MILP formulation that jointly optimizes 3D AP coordinates and azimuth orientations under a complete THz channel model, minimizing the number of nodes needed to guarantee user-level SINR coverage.
- An HPSO algorithm for real-time dynamic repositioning, with a composite fitness function that prioritizes human-density-weighted coverage while penalizing unnecessary physical displacement.
- A 4-tier cascading architecture that escalates from electronic beam steering to full spatial repositioning only when needed, significantly recovering user coverage

while minimizing total physical node displacement compared to alternative approaches.

II. Related Work

The THz band (0.1–10 THz) offers very large bandwidth for 6G, but suffers from high path loss, molecular absorption, and severe line-of-sight (LoS) blockage [**?**], [**?**], [**?**]. Human bodies can cause more than 20 dB attenuation [**?**], while pedestrian density and mobility strongly affect THz outage and SINR [**?**], [**?**]. These characteristics make static deployments vulnerable to spatiotemporally varying human blockage.

Existing AP placement methods primarily optimize the network before deployment. Baswade *et al.* [**?**] formulated base-station placement using mixed-integer optimization, while graph clustering for efficient mmWave LoS coverage was proposed by Alameh *et al.* [**?**]. Such approaches provide effective initial layouts but do not adapt the deployment after propagation conditions change. PSO and its variants have also been applied to wireless optimization, including 3D UAV-BS placement [**?**] and mmWave massive-MIMO beamforming [**?**].However, these approaches mainly focus on optimizing placement or communication parameters, rather than combining electronic adaptation with physical AP reconfiguration when human blockage changes. Physical mobility offers another means of adapting wireless coverage as user locations vary. Wireless platforms based on UAV can adjust their position in three dimensions to improve coverage and optimize their trajectories [**?**], [**?**]. Similarly, movable APs have been studied for indoor WLANs, where the AP locations are changed according to user distribution [**?**]. Movable antennas further exploit spatial channel variations by dynamically changing antenna positions [**?**]. Nevertheless, these approaches do not jointly consider indoor THz beam adaptation, vertical AP adjustment, and progressive full-3D repositioning based on human blockage.

Dynamic adaptation also requires information about human movement. Vision-based learning has been used to predict impending mmWave/THz blockages [**?**]. Specifically, 3D LiDAR point clouds have been extensively explored for high-resolution, indoor crowd counting and geometry tracking [**?**], [**?**]. By natively mapping spatial geometry, LiDAR provides a highly reliable foundation for contextual tracking [**?**], [**?**].

III. System Model and Problem Formulation

Designing indoor Terahertz (THz) networks requires strategic placement of APs to ensure reliable coverage. The positioning must account for the building's permanent structural layout as well as the random movement of people within it. The key notations used throughout this paper are given in Table **??**.

The indoor environment is discretized into a 3D volumetric grid. Let $\mathcal{U}$ denote the set of target user equipment locations requiring coverage, and let $\mathcal{A}$ denote the set of candidate AP locations. Unlike traditional 2D placement models, we explicitly optimize over the vertical Z-axis to accurately model signal propagation. Let the 3D coordinates of the user device

TABLE I: Glossary of Proposed Model

Notation	Definition
$\mathcal{U}, \mathcal{A}$	Set of target user locations and candidate AP locations
$d_{u,a}$	3D Euclidean distance between u and a (meters)
$d^{2D}_{u,a}, d^{2D}_{u,c}$	2D projected distances on the horizontal plane (meters)
ϕ_a	Azimuth pointing direction (orientation) of AP a
$\Delta\phi$	Angular deviation between user location and AP orientation
θ_{HPBW}	Half-Power (azimuth) Beamwidth of the directional THz antenna
θ_{el}	Fixed elevation beamwidth of the sector antenna
h_{human}	Average physical height of a human blocker
$z(c)$	Interpolated elevation of the signal beam at grid cell c
$W_a \in \{0,1\}$	1 if an AP is deployed at location a, 0 otherwise
$Y_{u,a} \in \{0,1\}$	1 if user u is associated with AP a, 0 otherwise
η	Target coverage threshold parameter (ratio, e.g., 0.90)
f	Operating frequency (Hz)
c	Speed of light (m/s)
$k(f)$	Molecular absorption coefficient at frequency f
$P_h(u)$	Empirical density of human presence at location u
$P_{\text{tx}}, P_{\text{rx}}$	Transmit and received signal power (dBm)
$G_{\text{tx}}, G_{\text{rx}}$	Transmit and receive antenna gains (dBi)
$G_{\text{max}}, G_{\text{min}}$	Main-lobe peak gain and side-lobe leakage floor (dBi)
SINR_{th}	Minimum required SINR in each target sub-region
N_0	System noise floor (linear scale)
L_w	Penetration penalty for static walls (dB)
L_h	Penetration penalty for human blockages (dB)
Inf	An arbitrarily large positive constant
$v_i^{(t)}, x_i^{(t)}$	Velocity and position of HPSO particle i at iteration t
w	Inertia weight (linearly decreasing from 0.9 to 0.4)
c_1, c_2	Cognitive and social learning factors
p_i^{best}	Personal best position of particle i
g^{best}	Global best position discovered by the swarm
F	Composite fitness score of a candidate AP configuration
R_{cell}	Ratio of grid cells meeting the SINR threshold
R_{density}	Ratio of covered human density to total density
B	Bonus awarded when target coverage is achieved
M	Normalized movement penalty for physical displacement
C_p	Anti-clustering penalty for AP spatial diversity
w_c, w_d	Coverage and density weights in fitness function
β, γ	Displacement and clustering penalty coefficients
$\mathcal{R}_{\text{prev}}, \mathcal{R}_{\text{new}}$	Sets of current and optimized AP positions
C_{ij}	Displacement cost to move AP r_i to new position r'_j
σ^*	Optimal AP-to-position assignment (Hungarian)

u be (x_u, y_u, z_u) and the candidate AP a be (x_a, y_a, z_a). The volumetric Euclidean distance $d_{u,a}$ is calculated as:

$$d_{u,a} = \sqrt{(x_u - x_a)^2 + (y_u - y_a)^2 + (z_u - z_a)^2} \quad (1)$$

where the user height z_u is fixed to a standard level, and the optimization algorithm selects an optimal AP mounting elevation z_a from a discrete set of candidate heights.

This 3D distance $d_{u,a}$ denotes the fundamental free-space propagation. The base Free Space Loss (FSL) for a signal operating at frequency f over distance $d_{u,a}$ is derived from the Friis transmission equation:

$$PL_{\text{FSL}}(d_{u,a}) = 20 \log_{10} \left(\frac{4\pi f d_{u,a}}{c} \right) \quad (2)$$

where c is the speed of light in a vacuum. The total deterministic (line-of-sight) path loss $PL_{\text{LoS}}(u, a)$ for a THz link further incorporates molecular absorption loss and static attenuation [**?**]:

$$PL_{\text{LoS}}(u, a) = PL_{\text{FSL}}(d_{u,a}) + PL_{\text{ABS}}(d_{u,a}) + L_{\text{static}} \quad (3)$$

where $PL_{\text{FSL}}(d_{u,a})$ represents the free-space path loss, L_{static} denotes the attenuation penalty from static infrastructure, and $PL_{\text{ABS}}(d_{u,a})$ accounts for atmospheric molecular absorption at high frequencies [**?**], calculated in dB as:

$$PL_{\text{ABS}}(d_{u,a}) = 4.343 \cdot k(f) \cdot d_{u,a} \quad (4)$$

with $k(f)$ representing the frequency-dependent molecular absorption coefficient.

Due to the highly directional nature of THz signals, line-of-sight (LoS) obstructions severely degrade link quality. We model two distinct types of blockages: static infrastructure and dynamic human mobility. For static obstacles, let $\mathcal{W}$ denote the set of walls within the environment. If the 3D line segment connecting u and a intersects a wall $w \in \mathcal{W}$, a material-specific penetration penalty L_w (e.g., concrete, wood, glass) is accumulated:

$$L_{\text{static}} = \sum_{w \in \mathcal{W}_{\text{intersect}}} L_w \quad (5)$$

Furthermore, THz signals suffer severe attenuation when occluded by human bodies [**?**]. When empirical human footfall datasets are available, we define an expected dynamic loss based on the density matrix $P_h(c) \in [0, 1]$ for the set of grid cells $\mathcal{C}_{u,a}$ intersected by the 2D projection of the link. A human only obstructs the signal if the elevation of the beam $z(c)$ as it passes through cell c drops below the standard human height h_{human}. This geometrical threshold is equivalent to the critical blockage distance limits derived from similar triangles in prior work on THz blockages for humans. Using linear interpolation based on 2D traversed distance $d_{u,c}^{2D}$ and total 2D distance $d_{u,a}^{2D}$, the beam elevation at cell c is given as:

$$z(c) = z_u + \frac{d_{u,c}^{2D}}{d_{u,a}^{2D}} (z_a - z_u) \quad (6)$$

The expected dynamic blockage loss is then defined using an indicator function $\mathbb{I}(\cdot)$, which equals 1 if the beam strikes the human body and 0 otherwise:

$$L_{\text{dynamic}} = \sum_{c \in \mathcal{C}_{u,a}} P_h(c) \cdot L_h \cdot \mathbb{I}\Big(z(c) \leq h_{\text{human}} \Big) \quad (7)$$

where L_h is the empirical penalty caused by human blockage.

To overcome extreme path loss, THz networks use highly directional antennas rather than omnidirectional antennas. We adopt a flat-top sector antenna model parameterized by the half-power beamwidth (HPBW), θ_{HPBW}, and the AP's mounting azimuth orientation ϕ_a. Although our model specifically analyzes the horizontal beamwidth, we assume that APs above the users ($z_a > z_u$) are optimally angled downward. This downward tilt ensures the main signal beam reaches the users directly. As a result, we can simplify our analysis by calculating the signal gain only in the horizontal plane.

Let $\phi_{u,a}$ denote the spatial angle of user u relative to AP a. The shortest angular deviation $\Delta\phi$ between the user's location and the AP's orientation, is calculated as:

$$\Delta\phi = \left| (\phi_{u,a} - \phi_a + 180^\circ) \bmod 360^\circ - 180^\circ \right| \quad (8)$$

This angular deviation ($\Delta\phi$) determines the transmit gain $G_{\text{tx}}(\Delta\phi)$ applied to the link:

$$G_{\text{tx}}(\Delta\phi) = \begin{cases} G_{\text{max}}, & \text{if } \Delta\phi \leq \frac{\theta_{\text{HPBW}}}{2} \\ G_{\text{min}}, & \text{otherwise} \end{cases} \quad (9)$$

where G_{max} is the gain in the main signal direction, and G_{min} represents the small amount of signal leakage in other directions. Since the total radiated energy remains constant, reducing the beamwidth (θ_{HPBW}) increases the main-lobe gain G_{max}. Using APs with narrower beamwidths increases the signal strength and allows communication over longer distances, but the coverage area becomes smaller. G_{min} is modeled as practically negligible, meaning the signal drops significantly outside the HPBW. Finally, the received power $P_{\text{rx}}^{(u,a)}$ (in linear scale) is obtained from the total path loss and the antenna gain as:

$$P_{\text{rx}}^{(u,a)} = 10^{\frac{P_{\text{tx}} + G_{\text{tx}}(\Delta\phi) + G_{\text{rx}} - PL(u,a) - 30}{10}} \quad (10)$$

IV. Proposed Work

Our objective is to ensure that every user within the indoor environment receives a minimum required SINR (SINR_{th}) using the fewest THz APs possible, even as human crowds dynamically reshape the blockage landscape. We address this in two stages: first, we formulate the static baseline placement as a MILP to compute a provably optimal initial deployment; second, we introduce an HPSO-based dynamic repositioning framework with a 4-tier cascading architecture to adapt the deployment in real time when crowd shifts degrade user coverage.

A. Context Acquisition from Human-Density Sensing

TeraMove uses spatial human-density information to capture changes in the blockage environment. In practice, this contextual data is obtained using 3D LiDAR sensors, which generate real-time topographical point clouds of the room. The raw 3D point cloud is divided into the exact volumetric grid cells (c) used by our formulation. By subtracting the static background, we isolate dynamic points corresponding to humans or moving furniture. The human density score $D_h(c) \in [0,1]$ for a specific vertical column is calculated as:

$$D_h(c) = \min\left(1, \frac{N_{\text{dyn}}(c)}{N_{\text{max}}}\right) \tag{11}$$

where $N_{\text{dyn}}(c)$ is the number of dynamic LiDAR points detected in the vertical column at cell c, and N_{max} is the expected maximum number of points for a single human body. In our evaluation, we generate the equivalent human-density map using an empirical pedestrian density model to evaluate coverage and trigger AP adaptation when needed

B. Static Baseline Placement via MILP

We formulate the baseline optimization problem as a Mixed-Integer Linear Programming (MILP) model. The primary objective is to minimize the total number of APs deployed, expressed by Eq. (**??**):

$$\min \sum_{a\in\mathcal{A}} W_a \tag{12}$$

Eq. (**??**) ensures that each target sub-region is associated with exactly one deployed AP. Eq. (**??**) mandates that a user u can only logically associate with a candidate location a if an AP is physically deployed there ($W_a = 1$).

$$\sum_{a\in\mathcal{A}} Y_{u,a} = 1, \quad \forall u \in \mathcal{U} \tag{13}$$

$$Y_{u,a} \le W_a, \quad \forall u \in \mathcal{U}, \forall a \in \mathcal{A} \tag{14}$$

In practical deployment scenarios where satisfying 100% coverage is physically or economically infeasible, Eq. (**??**) is relaxed to an inequality ($\sum_{a\in\mathcal{A}} Y_{u,a} \le 1$), and a global coverage constraint is introduced. Let $\eta \in (0,1]$ denote the required proportion of the total environment that must be served. To strictly enforce this spatial coverage ratio, the total number of successfully associated sub-regions must satisfy the following constraint:

$$\sum_{u\in\mathcal{U}} \sum_{a\in\mathcal{A}} Y_{u,a} \ge \eta|\mathcal{U}| \tag{15}$$

where $|\mathcal{U}|$ represents the total number of target user sub-regions in the partitioned grid.

The next constraint is to maintain the minimum SINR *i.e.*, threshold in each sub-region considering multi AP interference. Eq. (**??**) ensures that the minimum SINR is maintained in each sub-region:

$$\frac{P_{\text{rx}}^{(u,a)} + \text{Inf}\cdot(1-Y_{u,a})}{N_0 + \sum_{\substack{b\in\mathcal{A}\\ b\neq a}} P_{\text{rx}}^{(u,b)} W_b} \ge \text{SINR}_{\text{th}}, \quad \forall u \in \mathcal{U}, a \in \mathcal{A} \tag{16}$$

The Left Hand Side (LHS) of Eq. (**??**) is the ratio of received signal power from the AP to which the user is associated, to the sum of system noise N_0 and interference from other deployed APs. When the uth user is associated with an AP located at a, $Y_{u,a} = 1$. So, the term $\mathit{Inf}\cdot(1-Y_{u,a})$ goes to zero, and the LHS of Eq. (**??**) has to satisfy the constraint. Otherwise, if $Y_{u,a} = 0$, the term $\mathit{Inf}\cdot(1-Y_{u,a})$ takes on an arbitrarily large value. Because of this term, the numerator becomes extremely large and inherently satisfies the inequality, effectively ignoring the constraint for unassociated links [**?**]. The term $\mathit{Inf}\cdot(1-Y_{u,a})$ is needed because a single AP might not provide the minimum SINR to all sub-regions in the building simultaneously; this formulation ensures the overall feasibility of the MILP equations.

The above constraint can be linearized by cross-multiplying the denominator to the right-hand side. Since $P_{\text{rx}}^{(u,b)}$ is a precomputed constant for each link and W_b is a binary decision variable, their product $P_{\text{rx}}^{(u,b)} W_b$ is inherently linear. The linearized form of Eq.

$$P_{\text{rx}}^{(u,a)} + \text{Inf}\cdot(1-Y_{u,a}) \ge \text{SINR}_{\text{th}} \cdot \left(N_0 + \sum_{\substack{b\in\mathcal{A}\\ b\neq a}} P_{\text{rx}}^{(u,b)} W_b\right), \quad \forall u \in \mathcal{U}, a \in \mathcal{A} \tag{17}$$

This equivalent formulation ensures that the entire optimization problem remains within the MILP framework, enabling standard solvers to compute globally optimal AP deployments.

C. Dynamic AP Repositioning via HPSO

The MILP formulation produces a globally optimal AP layout for a given static environment. However, indoor human crowds are inherently dynamic, and a deployment optimized for one crowd configuration may become suboptimal minutes later. Rather than deploying additional APs for every possible crowd pattern, we propose dynamically repositioning the existing APs to track the evolving human density landscape.

To accomplish this, we employ a Hierarchical Particle Swarm Optimization (HPSO) algorithm built on the standard velocity-update formulation of [**?**], [**?**], with the hierarchy realized through the tiered activation of search dimensions described below rather than through evolutionary-state-based parameter adaptation. HPSO is a population-based metaheuristic where a swarm of candidate solutions (particles) explores the search space cooperatively. Each particle maintains a position vector x_i encoding the AP configuration parameters and a velocity vector v_i governing its exploration trajectory. The swarm evolves over T iterations according to the velocity update rule:

$$v_i^{(t+1)} = w\cdot v_i^{(t)} + c_1 r_1 (p_i^{\text{best}} - x_i^{(t)}) + c_2 r_2 (g^{\text{best}} - x_i^{(t)}) \tag{18}$$

where w is the inertia weight that linearly decreases from 0.9 to 0.4 over the course of optimization to transition from exploration to exploitation, c_1 and c_2 are cognitive and social learning factors (both set to 1.5), $r_1, r_2 \sim U(0,1)$ are random

coefficients, p_i^{best} is the personal best position of particle i, and g^{best} is the global best position discovered by the entire swarm. The position is subsequently updated as $x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)}$, with per-dimension velocity clamping to prevent divergence.

The fitness function that guides the swarm combines multiple objectives into a single composite score:

$$F = w_c \cdot R_{\text{cell}} + w_d \cdot R_{\text{density}} + B - \beta \cdot M - \gamma \cdot C_p \quad (19)$$

where R_{cell} is the ratio of grid cells meeting the SINR threshold, R_{density} is the ratio of covered human density to total density, B is a bonus awarded when target coverage is achieved, and M is a normalized movement penalty encouraging minimal physical displacement. C_p is an anti-clustering penalty that discourages APs from congregating too closely, formalized as:

$$C_p = \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} \max\left(0,\ 1 - \frac{\|r_i - r_j\|}{d_{\min}}\right) \quad (20)$$

where n is the number of deployed APs, r_i, r_j are their 3D positions, and $d_{\min}$ is a minimum desired inter-AP spacing (Table **??**). The weights $w_c = 0.4$ and $w_d = 0.6$ prioritize human-centric coverage, while $\beta = 0.25$ and $\gamma = 0.3$ govern the displacement and clustering penalties respectively.

1) 4-Tier Cascading Optimization Architecture: A key design principle of our dynamic framework is the minimization of physical AP displacement. Moving APs incurs practical costs: service interruption during relocation, mechanical wear, and potential re-calibration requirements. To address this, we introduce a 4-tier cascading optimization architecture that progressively unlocks more degrees of freedom only when simpler adjustments prove insufficient.

- **Tier 1 – Baseline Evaluation:** The system evaluates the current AP configuration against the updated human density map. If the density-weighted coverage already meets the target threshold ($\geq 90\%$), no optimization is triggered. This avoids unnecessary computation and physical displacement.
- **Tier 2 – Beam Parameter Optimization:** If coverage is insufficient, the HPSO optimizes only the azimuth orientation (ϕ_a) and beamwidth (α) of each AP. Since these are electronic adjustments requiring no physical movement, Tier 2 achieves coverage improvements at zero displacement cost.
- **Tier 3 – Height Adjustment:** If Tier 2 fails to restore adequate coverage, the optimizer additionally adjusts the vertical mounting elevation (z_a) of each AP from the set of candidate heights. This involves minor physical adjustment along the Z-axis only.
- **Tier 4 – Full Spatial Repositioning:** As a last resort, the HPSO unlocks the horizontal coordinates (x_a, y_a), enabling complete 3D relocation of APs. To prevent excessive displacement, one-third of the particle swarm is seeded near the current AP positions, allowing the optimizer to preferentially discover nearby solutions.

At each tier, the HPSO locks the velocity components of the dimensions not yet activated (e.g., in Tier 2, the x, y, and z velocity components are clamped to zero). This cascading structure ensures that the system always attempts the least disruptive adjustment first before escalating.

2) Displacement Minimization via Hungarian Algorithm: When the HPSO produces an optimized set of AP positions, a critical assignment problem arises: which *existing* AP should move to which *new* position? A naive assignment may result in unnecessarily large total displacement. To minimize the aggregate physical relocation distance, we formulate this as a classical assignment problem and solve it using the Hungarian algorithm [**?**].

Let $\mathcal{R}_{\text{prev}} = \{r_1, \ldots, r_n\}$ denote the n current AP positions and $\mathcal{R}_{\text{new}} = \{r'_1, \ldots, r'_n\}$ denote the n optimized positions. We construct a cost matrix $C \in \mathbb{R}^{n \times n}$ where each entry C_{ij} represents the 3D Euclidean displacement required to move existing AP r_i to new position r'_j:

$$C_{ij} = \sqrt{(x_i - x'_j)^2 + (y_i - y'_j)^2 + (z_i - z'_j)^2} \quad (21)$$

The Hungarian algorithm finds the optimal one-to-one assignment $\sigma^* : \{1, \ldots, n\} \to \{1, \ldots, n\}$ that minimizes the total displacement:

$$\sigma^* = \arg\min_{\sigma} \sum_{i=1}^{n} C_{i,\sigma(i)} \quad (22)$$

This assignment is computed in $O(n^3)$ time, which is negligible for the typical number of indoor APs ($n \leq 30$).

V. Experimental Setup and Simulation Results

For the experimental setup, we have considered an academic environment using a representative university department architectural layout. To capture the stochastic nature of dynamic human mobility, we used an empirical pedestrian mathematical model to generate spatial blockage probabilities. The simulation parameters are summarized in Table **??**.

We selected an SINR threshold of $\text{SINR}_{\text{th}} = 5$ dB as the primary evaluation criterion because it represents the minimum signal quality required for reliable THz link establishment in indoor environments. At 300 GHz, the extremely short wavelength makes signals highly susceptible to even minor blockage; a 5 dB threshold ensures that associated user subregions receive sufficient signal power to sustain basic data connectivity after accounting for molecular absorption, wall penetration losses, and multi-AP co-channel interference. This value aligns with the minimum SINR for low-order modulation schemes (e.g., QPSK) commonly used in THz feasibility studies [**?**].

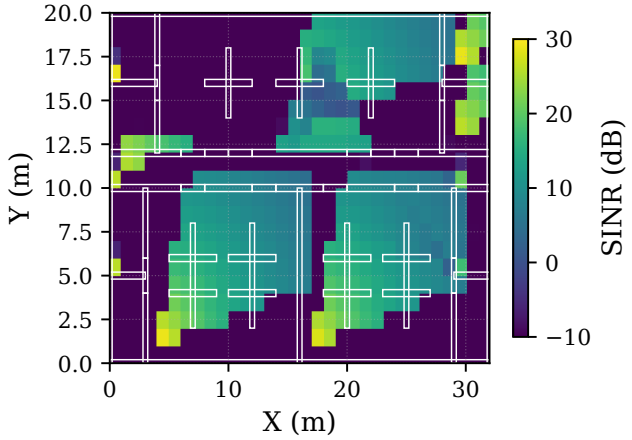


Fig. 1: SINR distribution achieving 100% coverage (SINR $\geq$ 5 dB).

TABLE II: Simulation Parameters

Parameter	Value
Environment Dimensions	32 m × 20 m × 3 m
Operating Frequency (f)	300 GHz
Transmit Power (P_{tx})	20 dBm
Main-Lobe Gain ($G_{\max}$)	30 dBi
Side-Lobe Leakage ($G_{\min}$)	Negligible ($\leq$ Noise Floor)
Evaluated Beamwidths (θ_{HPBW})	$30°, 45°, 60°, 75°, 90°, 120°$
Candidate Z-axis Elevations (z_a)	2.0 m to 3.0 m (in 0.2 m increments)
Azimuth Rotation Increments	$45°$ ($\pi/4$ rad, from North)
Target Coverage Threshold (η)	0.90 (90%), 0.95 (95%), 1.0 (100%)
Minimum Required SINR (SINR_{th})	5 dB (and 10 dB for strict testing)
System Noise Floor (N_0)	-94 dBm (Linear: 3.981×10^{-10} mW)
Solid Concrete Wall Penalty (L_w)	30 dB
Wooden Door/Cubicle Penalty (L_w)	20 dB
Glass Window Penalty (L_w)	15 dB
Static Cubicle/Desk Height	1.0 m
Human Blocker Penalty (L_h)	20 dB (height 1.6 m)

A. Static Baseline Deployment

We first evaluate the MILP formulation in a static environment. The solver identifies that 14 APs are needed to achieve 100% coverage at a minimum $\text{SINR}_{\text{th}} = 5$ dB. By evaluating candidate elevations in 0.2 m increments from 2.0 m to 3.0 m, the 3D optimization allows APs to transmit above 1.0 m cubicles. This 3D spatial optimization yields a 20% higher peak SINR compared to the constraint of restricting APs to a uniform height. Fig. **??** shows that this configuration meets the required SINR across all grid cells.

Further analysis shows that increasing the required SINR_{th} from 1 dB to 5 dB increases the AP count from 9 to 10 for 90% coverage. Regarding the antenna directionality, wider beamwidths ($90°$ to $120°$) require only 10 APs to achieve 90% coverage, whereas narrow beams ($60°$ to $75°$) require 15. In indoor scenarios, the broader reach of wider beams yields better overall coverage than the reach of narrow beams.

B. Tier-wise Coverage and Displacement Analysis

We tested our 4-tier cascading HPSO setup in a typical situation where humans block the signal, dropping the initial coverage to 63.6%. Fig. **??** shows how much each tier helps. Starting from a basic 14-AP layout dealing with human blockages, Tier 2 boosts coverage by an extra 26.5 percentage points (bringing it up to 90.1%) without moving any equipment at all (0.0 m). Tier 3 adds another 3.5 percentage points for just 3.1 m of movement, while Tier 4 brings in a final 3.3 percentage points at a cost of 10.7 m. As shown in Table **??**, these steps bring the total coverage back up to an effective 96.9%.

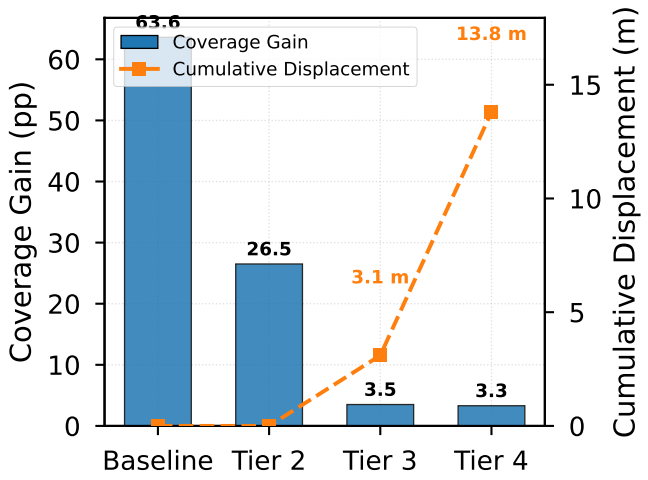


Fig. 2: Incremental coverage improvement and displacement cost at each optimization tier.

TABLE III: Tier-wise Coverage and Displacement Summary

Tier	Coverage Gain (pp)	Displacement (m)	Adjustment Type
Baseline	63.6% (Initial)	0.0	None
Tier 2	+26.5	0.0	Electronic (beam steering)
Tier 3	+3.5	3.1	Minor physical (Z-axis)
Tier 4	+3.3	10.7	Full 3D repositioning

The primary objective of the 4-tier approach is to prioritize the least disruptive adjustments. If coverage is already sufficient, Tier 1 retains the configuration to conserve computational and mechanical resources. When blockages cause significant coverage drops, Tier 2 executes the majority of the recovery. It restores 26.5 percentage points of the lost coverage—over 79% of the total improvement—at zero physical cost. This is achieved by bypassing THz blockages through electronic adjustment of beam direction and width. Practically, these Tier 2 adjustments occur in milliseconds via phased-array controls, making them highly suitable for real-time mitigation.

Tier 3 is preferred over Tier 4 because it only requires minor vertical adjustments, which consume significantly less time and energy compared to horizontal robotic relocation. Tier 4 is strictly reserved as a final measure when the blockage severity renders the current AP coordinates entirely unviable. On top of this, when we actually have to move devices in Tiers 3 and 4, we use the Hungarian assignment algorithm (explained in Section IV-D) instead of a basic nearest-neighbor approach.

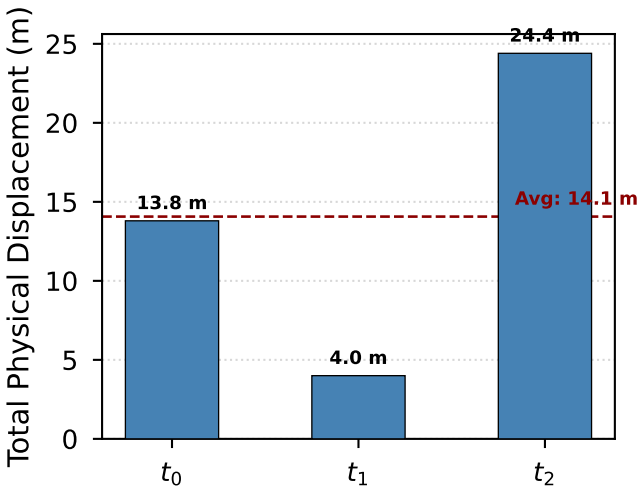


Fig. 3: Aggregate AP mechanical displacement per time step, highlighting minimal movement (4.0 m) required during t_1.

This cuts down the total movement distance by 8.6%, saving about 4.1 meters of overall travel every time we run the cycle.

C. Temporal Crowd Adaptation

To evaluate robustness under realistic temporal scenario we simulated three consecutive crowd shifts at discrete time steps (t_0, t_1, t_2), each representing a distinct mobility scenario drawn from empirical pedestrian mathematical model. At t_0 dense crowd clustering drops the coverage to 16.6%. At t_1, the crowd redistributes across rooms with less severe blockage, resulting in initial coverage of 80.2%. At t_2, a new dense gathering pattern emerges with different geometry, dropping coverage to 58.4%.

Fig. **??** shows the required AP displacement at each timestamp. At t_0, the 4-tier HPSO recovers coverage from 16.6% to 97.1%. During t_1, coverage is elevated from 80.2% to 95.6%. At t_2, HPSO significantly restores coverage from the 58.4% drop. Notably, as shown in Fig. **??**, the t_1 adjustment requires only minimal physical displacement, whereas t_0 and t_2 require more substantial relocation to adapt to the drastic environmental changes. This highlights that for less severe crowd shifts (e.g., t_1), Tier 2 beam steering accounts for the most of the recovery, handling blockages with minimal physical movement.

Fig. **??** presents the adaptation sequence at t_1. Starting from the t_0-optimized AP configuration (a) and the stable SINR map established after t_0 optimization (b), a new unpredicted crowd shift (c) disturbs these signals, degrading SINR coverage (d). The HPSO framework then repositions and re-angles the APs (e), resulting in fully restored SINR coverage (f). This sequence illustrates the framework's real time response to dynamic blockages.

D. Comparison with Alternative Approaches

We evaluate the HPSO against heuristic based algorithms : standard Particle Swarm Optimization (PSO) and a Genetic Algorithm (GA).We intentionally exclude combinatorial approaches, such as Greedy Set Cover, from the benchmarks because they discard MILP outcome of the AP's positional states and treat each time step as an new scenario and thereby failing to preserve the physical movement continuity required for router swarms. The unconstrained PSO serves as a baseline to isolate the benefits of the proposed hierarchical tier constraints. Baseline PSO optimizes the identical composite objective function Eq. (19) (including displacement penalty βM), but without the tier-clamping hierarchy, updating all 3D coordinates and beam parameters simultaneously. GA serves as a population-based benchmark to verify that improvements are not specific to the PSO framework.

The simulation parameters used are set as follows: Swarm size $S = 50$, maximum iterations $T = 20$, which were evaluated over 30 Monte Carlo runs with independent random seeds.

Fig. **??** illustrates network coverage as human density increases. By incorporating footfall density into the fitness function, HPSO maintains over 88

The hierarchical structure reduces mechanical movement by prioritizing electronic beam-steering before starting physical relocation. Under unconstrained optimization, standard PSO updates all spatial coordinates and beam parameters simultaneously which results in a total displacement of over 450 m across the simulated scenarios. In contrast, the proposed hierarchical model limits total physical displacement to 42.2 m, achieving a 90.6

Fig. **??** compares convergence rates. By restricting physical movement variables in early tiers and allowing the swarm to first explore the beam-steering solution space (Tier 2), HPSO reaches optimal fitness faster than both standard PSO and GA. Standard PSO explores all dimensions simultaneously, which leads to slower convergence and is highly sensitive to local optima. GA exhibits the slowest convergence because it relies on randomized search processes, which are less efficient than velocity-based updates for continuous optimization in high-dimensional spaces.

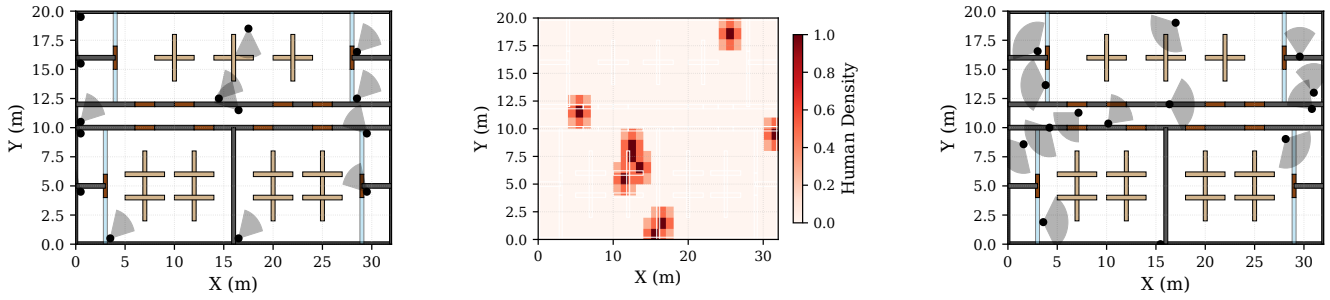
TABLE IV: Fitness Weight Sensitivity Analysis

Configuration Focus	w_c	w_d	β	γ	Avg. Cov. (%)	Total Disp. (m)
A: Moderate Penalties	0.4	0.6	0.15	0.3	91.0	113.2
B: High Density Focus	0.3	0.7	0.15	0.3	96.2	117.1
C: Low Clustering Penalty	0.4	0.6	0.15	0.1	91.6	104.7
D: High Movement Penalty	**0.4**	**0.6**	**0.25**	**0.3**	**95.2**	**42.2**

E. HPSO Hyperparameter Sensitivity

Standard PSO hyperparameters (inertia weight w, learning factors c_1, c_2) follow established guidelines [**?**], [**?**]. To adjust the domain-specific fitness weights $(w_c, w_d, \beta, \gamma)$, we performed a sensitivity analysis across continuous multiple steps tracking $(t_0 \rightarrow t_2)$ to assess the balance between coverage restoration and physical displacement. Table **??** displays four configurations demonstrating the influence of fitness weights on average dynamic coverage and total swarm displacement.

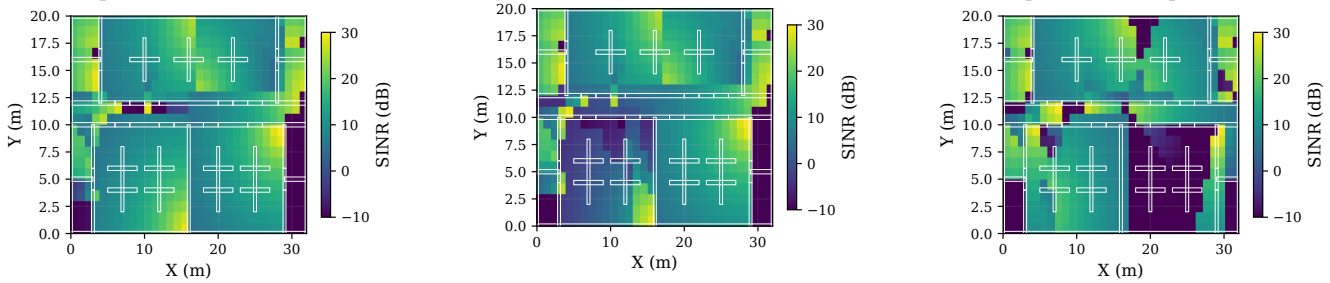
The configurations demonstrate divergent behaviors during continuous dynamic tracking. Configuration B heavily prior-



(a) Step 1: Baseline AP Positions

(c) Step 3: New Human Density Shifts

(e) Step 5: HPSO Repositions APs



(b) Step 2: Established Stable SINR

(d) Step 4: Blockage Degrades SINR

(f) Step 6: Restored SINR Coverage

Fig. 4: Chronological sequence of the adaptation at t_1. The sequence begins with the t_0-optimized AP configuration (a) and its stable SINR (b). A new crowd shift (c) disturbs these signals (d). HPSO repositions and re-angles APs (e) to restore coverage (f).

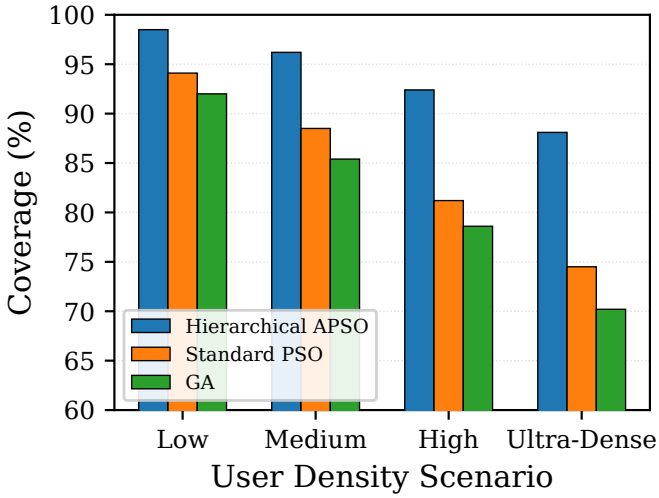


Fig. 5: HPSO maintains high coverage under high-density blockage; standard PSO degrades significantly.

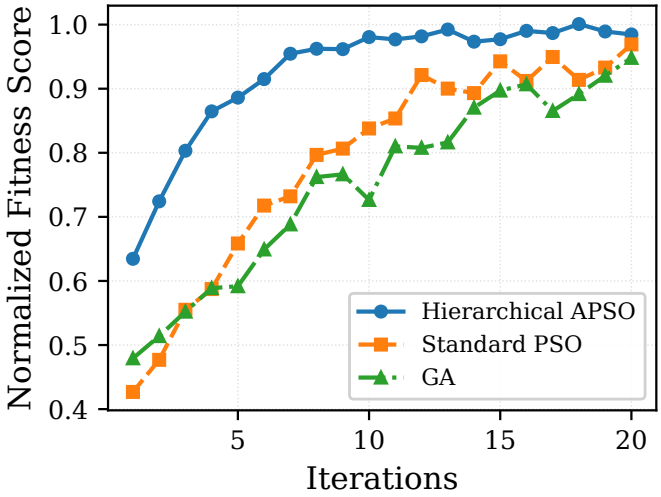


Fig. 6: Convergence comparison: HPSO achieves optimal fitness faster than standard PSO and GA.

itizes density ($w_d = 0.7$), leading to excessive physical displacement (117.1 m) as the swarm overfits to temporary crowd clusters and needlessly chases crowd dispersal. Furthermore, relaxing the anti-clustering penalty ($\gamma = 0.1$) in Configuration C increases the likelihood of AP interference due to proximity, degrading dynamic coverage to 91.6%. Configuration A provides a moderate movement penalty ($\beta = 0.15$), but this leads to unnecessary spatial exploration, incurring a high dynamic displacement (113.2 m) without yielding optimal coverage (91.0%).

Conversely, **Configuration D** applies a high movement penalty ($\beta = 0.25$). The dynamic simulations reveal that this high penalty successfully forces the swarm to rely on efficient electronic beam steering.Configuration D prevents inconsistent physical relocation, allowing APs to settle into stable and efficient positions. Consequently, it minimizes total

dynamic displacement to just 42.2 m while sustaining an excellent 95.2% average dynamic coverage. We therefore adopt **Configuration D** as our default parameters across multi-step evaluations (Section V-D) because it provides the optimal balance of high coverage and minimal mechanical movement.

VI. Discussion

The results presented in this work demonstrate the efficacy of the 4-tier cascading HPSO framework, but the practical deployment of this system can be realized through two distinct operational paradigms: real-time reactive adaptation or scheduled time-based adaptation.

In a **real-time adaptation model**, the HPSO framework operates as an active control loop integrated directly into the building's IoT infrastructure. This requires real-time crowd sensing mechanisms, such as camera-based tracking, Wi-Fi probe requests, or infrared occupancy sensors, to continuously generate the human density spatial matrix. Whenever the sensors detect a severe crowd shift that drops the network below the 90% coverage threshold, the HPSO triggers immediately. By relying heavily on Tier 2 electronic beam steering, it corrects the geometric blockages instantly without physical motion, only unlocking the robotic mounting tracks (Tiers 3 and 4) when the real-time blockages are insurmountable via software.

Alternatively, the framework can be deployed as a **time-based scheduled system**. Human mobility in large indoor environments such as universities or corporate offices often follows highly predictable temporal routines (e.g., morning arrivals, lunch hours, or class changeovers). Instead of requiring expensive real-time tracking sensors, network operators can use historical footfall data to predict density blockages at specific hours of the day. The HPSO algorithm would pre-calculate the optimal layout for these routine crowd shifts. At the scheduled time, the physical APs simply execute the required robotic repositioning and beam steering to match the pre-solved optimal configuration, ensuring proactive coverage without the overhead of live environmental scanning.

From a **mechanical infrastructure** perspective, the physical realization of Tiers 3 and 4 requires motorized mounting systems for the APs. Tier 3 height adjustments can be implemented using linear actuators or servo-driven vertical rail mounts attached to ceiling tracks, enabling controlled Z-axis displacement of a few centimeters to meters. Tier 4 full 3D repositioning demands a more sophisticated infrastructure: ceiling-mounted robotic rail systems with bidirectional tracks allow APs to traverse the horizontal plane, while integrated cabling harnesses, such as power-over-Ethernet (PoE) flexible cable chains or retractable cable reels, maintain uninterrupted power and backhaul connectivity during movement. The total cable management complexity scales linearly with the number of mobile APs and the maximum rail length. Importantly, the Hungarian algorithm-based displacement minimization (Section V-G) directly reduces the mechanical wear on these systems by ensuring each reconfiguration cycle uses the minimum total travel distance, extending the operational lifetime of the motorized mounts and reducing maintenance costs.

VII. Conclusion

This paper presents a comprehensive framework for 6G Terahertz (THz) Access Point deployment, bridging the gap between theoretical limits and unpredictable indoor environments. We initially demonstrated that optimizing AP height and beam orientation via a 3D MILP formulation significantly reduces the required number of APs. However, because static models inevitably degrade when moving human crowds block fragile high-frequency signals, we introduced a continuous Human Density tracking model to adapt to spatial blockage in real-time.

To overcome these blockages, we developed a 4-tier Hierarchical Particle Swarm Optimization (HPSO) architecture. By hierarchically escalating from fast, zero-movement electronic beam steering up to full 3D robotic repositioning only when absolutely necessary, our framework successfully restores network coverage even during dense crowd shifts. Reinforced by the Hungarian algorithm to minimize mechanical displacement, this research proves that the reliable THz networks of tomorrow will not be achieved by senselessly installing more APs, but by intelligently and dynamically moving with the crowds they serve.

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